Khatam: Proximity Gaps For Multilinear Evaluation For All Linear Codes

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Abstract. Two techniques have recently emerged in the construction of Succinct Non-Interactive Arguments of Knowledge (SNARKs) that yield extremely fast provers; The use of multilinear (instead of univariate) polynomial commitment schemes (PCS) and the construction of practical SNARKs from error-correcting codes. Recently, BaseFold (Crypto 2024) introduced a family of SNARKs that combine these two techniques, thereby achieving a better trade-off between prover time and verifier costs than prior work. Despite its impressive overall efficiency, BaseFold suffered from larger proof sizes than its univariate counterparts, due to unproven claims about linear codes, which were not relevant in the univariate setting.

This work closes this gap by proving a new fact about linear codes – that if π_L, π_R are two vectors in $\mathbb{F}^n$ and if $\pi_L + r\pi_R$ is close to a codeword in C, then π_L, π_R and $(\pi_L + r\pi_R)$ all agree with codewords at positions in the same set $S \subset [n]$, except with negligible probability over $r \leftarrow \mathbb{F}$. Our result holds as long as $|S| > ((1 - \Delta_C + \epsilon)^{1/3} + \eta)n$, for $\epsilon, \eta \in [0, 1]$ and with failure probability smaller than $\frac{3}{\epsilon\eta|\mathbb{F}|}$, where Δ_C is the minimum distance of the code. Importantly, our results extend to any finite field and any linear code, and result in a $2\times$ reduction in proof size compared to BaseFold and other state-of-the-art field-agnostic, hash-based SNARKs.

1 Introduction

In a Succinct Non-Interactive Argument of Knowledge (SNARK), a powerful prover seeks to convince a resource constrained verifier that it posesses a witness $w \in \{0, 1\}^*$ that satisfies a public circuit, C. SNARKs built from error-correcting codes (e.g. [1, 2, 6, 10, 13, 20]) have extremely efficient provers and fast verifiers, and are currently the backbone of several real-world systems (e.g. from Succinct Labs, StarkWare, Polygon[1]. However, a major drawback of hash-based SNARKs is their large proof size, which costs tens to thousands of kilobytes (in contrast to elliptic curve based SNARKs such as Groth16 [14], which have proof sizes that are only a few bytes).

[1] https://www.succinct.xyz/, https://starkware.co/, https://polygon.technology/

Proximity Gaps The seminal work of FRI [6] introduced the first practical hash-based SNARK. It's security relies on a fact about linear codes, referred to as "proximity gaps", which can be roughly stated as follows. If a random linear combination of two vectors is "close" in Hamming distance to a codeword, then with high probability, the two individual vectors also are and they agree with their respective (closest) codewords in the same positions as each other. But what distance from the code does the linear combination have to be for this property to hold? This is the question behind a series of papers (e.g. [5, 8]), which aim to prove that this distance does not have to be too close. Checking that a vector is close to a code is expensive in terms of communication between the prover and verifier, which ultimately leads to larger proofs. Relaxing the distance leads to more efficient schemes, with smaller proofs. It was eventually proven in [8] for Reed-Solomon codes for the relaxed distance of $\sqrt{1-\Delta_C}$ in relative Hamming distance. The authors of [8] conjectured that it may be true even as far as $1-\Delta_C$. The Ethereum foundation has put out a prize worth a million dollars to solve this conjecture[2] (specifically for the setting of multilinear polynomials, which we discuss next).

Proximity Gaps For Multilinear Evaluation BaseFold [20] generalized FRI [6] from univariate to multilinear polynomials and from Reed-Solomon codes to more general family of codes. Proving the soundness of BaseFold requires the following, slightly stronger property. If the linear combination of two vectors is "close" in Hamming distance to a codeword, then 1) the individual vectors both are and their agreement positions are consistent and 2) the random linear combination is "close" to the codeword we expect (we will discuss what we mean by this in detail later). Since multilinear SNARKs have such fast provers and verifiers, it became an important goal for the community to prove this stronger version of proximity gaps, so that these faster schemes could match the proof sizes of their univariate counterparts. For instance, WHIR [2], a polynomial commitment scheme which iterates on BaseFold, formalized this stronger notion of proximity gaps, naming it "mutual correlated agreement". Haböck then proved in [16] that mutual correlated agreement holds for Reed-Solomon codes and with the same relaxed distance as in the univariate setting.

Field-Agnostic SNARKs In this paper, we prove proximity gaps for multilinear evaluation for any linear code, so that we can maintain small proof sizes even for SNARKs defined over codes other than Reed-Solomon. While Reed-Solomon codes yield better proof sizes, they are only efficient for applications that can be easily arithmetized over FFT-friendly fields. By proving multilinear proximity gaps for general linear codes, we improve the efficiency of a broader class of applications. As an example, the field-agnostic code family, Random Foldable Codes [20] match the prover efficiency of Reed-Solomon codes and can prove SNARKs defined over fields of small characteristic (e.g. $\mathbb{F}_{257}$), something that Reed-Solomon codes cannot do. Random Foldable Codes are also a building

[2] https://proximityprize.org/

block in a recent linear-time encodable, field-agnostic SNARK [4]. The results in this paper meaningfully lower the communication complexity of such schemes in the interactive setting, which translates to 2× smaller proofs when compiled into SNARKs via Fiat-Shamir.

Other Applications In addition to being field agnostic, our results yield SNARKs with a much smaller soundness error than in the Reed-Solomon case. This enables SNARKs over smaller fields, which leads to faster prover times. In that same vein, recent work [12] uses mutual correlated agreement as a black-box to prove round-by-round soundness of univariate FRI. In that paper, they instantiate their result with the bound from [16]. If they instead use this result, they can achieve a much lower soundness error.

Furthermore, we adapt a definition from DeepFRI [7] to the multilinear setting, which allows for some loss in distance between rounds. By tuning the loss parameter, we are able to increase the proof size by lowering the soundness error, which in turn allows for the prover to use a smaller, cheaper, finite field without compromising on security.

1.1 Our Results

Before describing our contributions, we give a more detailed description of proximity gaps for multilinear evaluation, which states the following. Let $n \in \mathbb{N}, \mathbf{v}_1, \mathbf{v}_2 \in \mathbb{F}^n, r \in \mathbb{F}$ and $\mathbf{v}_r = \mathbf{v}_1 + r\mathbf{v}_2$. If $\mathbf{v}_r$ is close to a codeword, then with high probability over choice of $r \leftarrow \mathbb{F}$, $\mathbf{v}_1, \mathbf{v}_2$ are close in Hamming distance to codewords c_1, c_2. Furthermore, $\mathbf{v}_1 - c_1$ and $\mathbf{v}_2 - c_2$ are equal to 0 in the same positions (correlated agreement). Finally, $\mathbf{v}_r$ is close to $c^* = c_1 + rc_2$. It turns out that this last property is true as long as $\mathbf{v}_r - c^*$ is zero in the same positions that $\mathbf{v}_1 - c_1$ and $\mathbf{v}_2 - c_2$. In fact, this is the inspiration behind the name "mutual correlated agreement", proposed by [2].

In this work, we prove a more generalized version of the above, which we adapt from Deep-FRI [7]. Instead of mandating that $\mathbf{v}_1 - c_1, \mathbf{v}_2 - c_2$, and $\mathbf{v}_r - c^*$ are all 0 in the same locations, we relax this so that the zero positions of $\mathbf{v}_1 - c_1$ and $\mathbf{v}_2 - c_2$ are a large subset of the zero positions of $\mathbf{v}_r - c^*$. In particular, and following the notation of Deep-FRI [7], we introduce a parameter $\eta \in [0, 1]$, and mandate that the set of zero positions of $\mathbf{v}_1 - c_1$ and $\mathbf{v}_2 - c_2$ has at most ηn fewer elements than the zero positions of $\mathbf{v}_r - c^*$. We summarize our main result with the following theorem.

Theorem 1 (Relaxed Mutual Correlated Agreement (Informal)). *Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{F}^n$, let C be a linear code and denote by Δ_C the relative minimum Hamming distance of C. Let $r \leftarrow \mathbb{F}$ be uniformly sampled from $\mathbb{F}$ and suppose that there exists a large set S, with $|S| > [(1 - \Delta_C + \epsilon)^{1/3}] \cdot n$ such that S is the zero-positions of $\mathbf{v}_1 + r\mathbf{v}_2 - c^*$, where c^* is it's closest codeword. Then, with probability $\frac{3}{\eta \cdot \epsilon \cdot \mathbb{F}}$, there exists $S' \subset S$ with $|S'| > |S| - \eta n$, such that the zero positions of $\mathbf{v}_1 - c_1, \mathbf{v}_2 - c_2$ lie in S'.*

Remark 1. We can recover the definition of mutual correlated agreement from WHIR [2] from Theorem 1 by setting $\eta = 1/n$ and $|S| = |S'|$.

Next, we will use the above theorem to improve the soundness of the BaseFold Interactive Oracle Proof of Proximity (IOPP).

Theorem 2 (BaseFold Soundness For Evaluation At A Random Point (Informal)). *Let $d = \log_2(n)$ be the number of rounds in the IOP, let $\eta, \epsilon \in \{0,1\}$ and let $\lambda \in \{0,1\}$ be a security parameter. A verifier that makes $\lambda / \log_2((1-\Delta_C+\epsilon)^{1/3}) - d\eta)$ queries will accept a malicious proof with probability less than $2^{-\lambda} + \frac{3d}{\eta\epsilon|\mathbb{F}|}$.*

Applications We demonstrate the general applications of Theorems 1 and 2, including the resulting efficiency gains to BaseFold and other multilinear polynomial commitment schemes. Compared to the original proof size of BaseFold, the new proof size is approximately $2\times$ smaller. This $2\times$ reduction is highly significant for blockchain applications, which are acutely sensitive to overhead from proof size. For context regarding the value of such optimizations, the Ethereum Foundation currently offers a 1 million bounty for solving mutual correlated agreement up to capacity—a milestone that would achieve a comparable 2x reduction over state-of-the-art Reed-Solomon schemes.

Furthermore, because of our relaxed version of mutual correlated agreement, we can tune η to achieve smaller soundness errors than what exists in other works (allowing SNARKs over smaller, more efficient finite fields). We demonstrate this tradeoff in Section 5.

Future Work Theorem 1 has applications beyond the soundness analysis of BaseFold. Setting $\eta = 1/n$ we recover mutual correlated agreement from [2], which means this result could be used to prove soundness of WHIR in the field-agnostic setting. This should enable Random Foldable Codes over WHIR with reasonable proof sizes, which would improve the verifier efficieny of field-agnostic schemes. Furthermore, these results can be used to improve the soundness error / communication complexity of Brakedown [13], Blaze [10] and other multilinear polynomial commitment schemes. Finally, while we only prove the simple definition of soundness here, it is straightforward to use this result to prove knowledge soundness of BaseFold, following the techniques described in [20].

1.2 Technical Overview

Proximity Gap For Multilinear Evaluation We now give a high level overview of the proof. We would like to bound the size of a set $\mathsf{BC} \subset \mathbb{F}$ of "bad" elements, where for all $r \in \mathsf{BC}$, there exists a large set $S \subset [n]$, and codewords $c_L, c_R \in C$, where

$$\pi_L[S] + r\pi_R[S] = c_L[S] + rc_R[S], \tag{1}$$

but there is no large set $S' \subset S$ such that

$$\pi_L[S'] = c_L[S'], \pi_R[S'] = c_R[S'], \tag{2}$$

where $\pi_L[S]$ denotes the vector $(\pi_L[i] : i \in S)$ (resp. for π_R) and $C[S] = \{c[S] : c \in C\}$. First, suppose that there is no large set $S' \subset S$ satisfying Equation 2. By linearity of C, this implies that there is only one $r \in \mathbb{F}$ satisfying Equation 1, i.e. satisfying $\pi_L[S] + r\pi_R[S] \in C[S]$. This already gives a naive upper bound on the size BC, as it is *at most* the number of subsets of $[n]$ that are larger than $|S|$, which can be computed by taking the sum of binomial coefficients (and which is bounded by 2^n). However, this is clearly insufficient as we would like $|\mathsf{BC}|/|\mathbb{F}|$ to be negligible.

Our strategy is as follows. Denote by S the set of "bad sets" which satisfy Equations 1, 2 with respect to some codeword pair $(c_L, c_R) \in C \times C$ and some field element $r \in \mathbb{F}$. As mentioned in the previous paragraph, for any $S \in \mathsf{S}$, there is at most one $r \in \mathsf{BC}$ satisfying Equations 1, 2. Using this fact, we show that $|\mathsf{BC}| \leq |\mathsf{S}|$ and going forward, we focus our attention on bounding the size of S. Next, we observe that if $S_1, S_2 \in \mathsf{S}$ and $|S_1 \cap S_2| > (1 - \Delta_C)n$ then there exists some $c_L, c_R \in C \times C$ such that

$$\pi_L[S_1] + r_1\pi_R[S_1] = c_L[S_1] + r_1 c_R[S_1],$$

$$\pi_L[S_2] + r_2\pi_R[S_2] = c_L[S_2] + r_2 c_R[S_2],$$

for some $r_1, r_2 \in \mathbb{F}$. Furthermore, we will show that any set $S_3 \in \mathsf{S}$ such that $|S_1 \cap S_2 \cap S_3| > (1 - \Delta_C)n$ also satisfies

$$\pi_L[S_3] + r_3\pi_R[S_3] = c_L[S_3] + r_3 c_R[S_3],$$

for some $r_3 \in \mathbb{F}$. But on the other hand, we will show that there can't be more than $\frac{1}{\eta}$ sets that satisfy Equations 1, 2 according to c_L, c_R. Therefore for every pair of sets, $S_1, S_2 \in \mathsf{S}$, there are at most $\frac{1}{\eta}$ sets that intersect with $S_1 \cap S_2$. Since there are at most $\binom{|\mathsf{S}|}{2}$ pairs of sets in S, there are at most $\binom{|\mathsf{S}|}{2}\frac{1}{\eta}$ triples of sets in S that have large three-way intersection. Finally we show that because S has only a relatively small number of triples with large three-way intersection, S must be smaller than $\frac{3}{\eta\epsilon}$.

BaseFold Soundness. Next, we describe how to use the above result to prove soundness of BaseFold. For the sake of this summary, we describe a simplified version of the protocol where the prover sends full messages to the verifier and operates over only one round. First, the prover sends a vector $\mathbf{v}$ and the verifier responds with a random field element $r \leftarrow \mathbb{F}$. Next, the prover decomposes $\mathbf{v}$ into two vectors $\mathbf{v}_L, \mathbf{v}_R$ and sends back $\mathbf{w} = \mathbf{v}_L + r\mathbf{v}_R$. The verifier will do a probabilistic check that there exists some large set $S \subset [n]$ and codeword $c^* \in C$ such that $\mathbf{w}[S] = c^*[S]$. We want to show that this gives a high probability guarantee that $\mathbf{v}_L[S'] = c_L[S'], \mathbf{v}_R[S'] = c_R[S']$, for a large subset $S' \subset S$ and codewords $c_L, c_R \in C$ such that $c_L + rc_R = c^*$.

We will show that the following two events happen with negligible probability. In the first event, a malicious prover commits to $\mathbf{v}$ such that $\mathbf{v}_L, \mathbf{v}_R$ are far from any codeword, but then the prover gets lucky and receives verifier challenge r such that $\mathbf{w} = \mathbf{v}_L + r\mathbf{v}_R$ is close to a codeword. The result from DeepFRI [7]

already shows that this happens with negligible probability. In the second event, a malicious prover sends $\mathbf{v}_L, \mathbf{v}_R$ which agree with c_L, c_R on some large set S, but after recieving the verifier challenge, r, manages to send $\mathbf{w}$ which is close to some other codeword that does not agree with $c_L + rc_R$ on S. Using Theorem 1, we show that this event only happens with negligible probability.

1.3 Related Work

Prior Work There are several papers [5, 7, 8] that analyze the communication complexity of FRI [6] using regular proximity gaps, which is only useful in the univariate setting. In "Worst Case To Average Case Reduction For Distance to a Linear Code" [5], the authors improve upon the original FRI paper by proving that $|\mathsf{BC}|$ is small for agreement sets, S, where $|S| \geq (1 - \Delta_C)^{1/4}n$ and Deep-FRI [7] improves this to $|S| \geq (1-\Delta_C+\epsilon)^{1/3}n$. Deep-FRI additionally introduces a modification of the FRI protocol which further reduces the number of verifier queries, at the cost of some (slight) prover overhead. Finally, in "Proximity Gaps of Reed-Solomon Codes" [8], the authors use a list-decoding algorithm for Reed-Solomon codes to prove correlated agreement for $|S| \geq \sqrt{1 - \Delta_C + \epsilon}n$, but (as mentioned earlier with regards to [16]), this result is only meaningful with a field that is at least quadratic in the instance size and with a suitably small rate, both of which impact prover time.

Concurrent Work There were a few concurrent and independent works that address the same problem as the one in this paper. In "BaseFold In The List Decoding Regime" [16], Haböck proves a similar statement, but where the size of the minimum agreement set is $|S| > \sqrt{1 - \Delta_C + \epsilon}$ and where the number of bad elements, BC is in $O(n^2)$, where $n \in \mathbb{N}$ is the number of elements in the codeword. Furthermore, their result only applies to Reed-Solomon codes. In other concurrent work, titled "Linear Proximity Gap for Linear Codes within the 1.5 Johnson Bound" [11], they address the same problem that we do and was originally released around the same time. At first, their result put the number of bad elements, BC in $O(n/\epsilon^3)$. They later took inspiration from an earlier draft of this work and improved the bound to a (slightly) better bound than what we present here (around n/ϵ versus $3/\epsilon$). We choose to keep the slightly worse bound in favor of a simpler proof. In Deep-Fold [15], Guo et al adapts Deep-FRI [7] to Basefold, and in that setting also proves the bound for $|S| > \sqrt{1 - \Delta_C}$, but this only applies to Reed-Solomon codes and only to the DeepFRI [7] Protocol, which is a variant of FRI with slightly more overhead. WHIR [2] was the first paper to give a name ("mutual correlated agreement") to the problem originally presented in BaseFold of adapting proximity gaps for multilinear polynomials. They present an improved result in the unique decoding regime with similarly low soundness error of n/ϵ for $|\mathsf{S}| > (1 - \Delta_C/2)$. However, this result only holds for Reed-Solomon codes.

Subsequent Work Since its initial appearance as a preprint in November 2024, the techniques and results of this work have found applications in several subsequent papers:

– "Query-Optimal IOPPs for Linear-Time Encodable Codes" [3] (Eurocrypt 2026) generalizes the techniques introduced herein to a broader setting (see Theorem 3.11).
– The security analysis presented in [9] draws directly upon the methodologies developed in this work [3]

Furthermore, our results offer immediate improvements to other recent constructions in the literature. For example, in "Improved Round-by-round Soundness IOPs via Reed-Muller Codes" [18] (FOCS 2025), the authors derive proximity gaps for Reed-Muller codes. Because our framework applies to arbitrary linear codes, including Reed-Muller codes, incorporating our bounds yields a direct improvement to their proximity gaps. Finally, a recent construction adapts the field-agnostic code family, random foldable codes, to the setting of Galois rings [19]. To build their protocol, the authors provide a proof that proximity gaps hold within the unique decoding radius over Galois rings. Applying our generalized results to their setting would bypass this requirement, reducing their proof size by a factor of approximately two.

2 Preliminaries

2.1 Notation

$\mathbb{F}$ is a finite field and $\mathbb{F}^* = \mathbb{F} \setminus \{0\}$. For $n \in \mathbb{Z}$. denote by $[n]$ the set $[0, n-1] \cap \mathbb{Z}$. For a set $S \subseteq [n]$, $\mathsf{even}(S)$ is the set of even integers in S, and $\mathsf{odd}(S)$ is the set of odd integers in S. For $q \in \mathbb{N}$, $S/q = \{s/q : s \in S\}, S + q = \{s + q : s \in S\}$, etc. 2^S is the power set of S. For $x \in \mathbb{F}, r \in \mathbb{N}$, $x^{||r}$ is the string obtained by concatenating x to itself r times. For function $f : S \to S$, and $n \in \mathbb{N}$, $f^{\circ n}$ is the composition of f with itself n times.

2.2 Vectors

Definition 1 (Restriction of Codeword). *For $n \in \mathbb{N}$, a vector $\mathbf{v} \in \mathbb{F}^n$, the restriction of $\mathbf{v}$ to set $S \subset [n]$ is subvector $(c[i] : i \in S)$. We denote the restriction by $c[S]$.*

Definition 2 (Hadamard Product). *Let $n \in \mathbb{N}$ and let $v_1, v_2 \in \mathbb{F}^n$. Then the hadamard product of v_1 and v_2, denoted by $v_1 \circ v_2$, is equal to the following vector.*

$$v_1 \circ v_2 = (v_1[i] \cdot v_2[i] : i \in [n]).$$

2.3 Binary Trees

In this section, we will use the following notation. For $n \in \mathbb{N}$, and $T = (V, E)$ with n leaves, set $d = \log_2(n)$ and denote by $V[i]$ the ith layer of vertices

[3] See page 11 in [9].

as the set of vertices with i-length paths to the root. For $i \in [\log_2(n)], v \in V[i]$, we denote by $\mathsf{children}_T(v)$ the two neighbors of v in layer $V[i+1]$ and similarly for $w \in \mathsf{children}_T(v)$, v is equal to $\mathsf{parent}_T(w)$. For $w \in \mathsf{children}_T(v)$, we denote by $\mathsf{sibling}_T(v)$ the set $\mathsf{children}_T(v) \setminus w$. For a set $S \subset V[i]$, we will use the notation $\mathsf{parent}_T(S)$ to be the set $\bigcup_{v \in S} \mathsf{parent}_T(v)$ and similarly for $\mathsf{sibling}_T(S), \mathsf{children}_T(S)$.

Definition 3 (Direct Path). *Let $n \in \mathbb{N}$, let T be a binary tree with n leaves, set $d := \log_2(n)$ and let $v \in V[d]$. $P_T(v)$ is the unique path from v to root.*

Definition 4 (Augmented Path). *Let $n \in \mathbb{N}$, let T be a binary tree with n leaves, let $d = \log_2(n)$, and let $v \in V[d]$. $P'_T(v)$, is defined as follows.*

$$P'_T(v) = P_T(v) \cup \bigcup_{w \in P_T(v)} \mathsf{sibling}_T(w).$$

2.4 Coding Theory

Definition 5 (Linear Code). *A linear error-correcting code with message length k and codeword length n is an injective mapping from $\mathbb{F}^k$ to a linear subspace $C \subseteq \mathbb{F}^n$. C is associated with a generator matrix, $G \in \mathbb{F}^{k \times n}$ such that the rows of G are a basis of C and the encoding of a vector $\mathbf{v} \in \mathbb{F}^k$ is $\mathbf{v} \cdot G$. The minimum Hamming distance of a code is the minimum on the number of different entries between any two different codewords $c_1, c_2 \in C$. If C has a minimum distance $d \in [n]$, we say that C is an $[n, k, d]$ code and use Δ_C to denote d/n—the relative minimum distance.*

Definition 6 (Pairwise distance). *Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2 \in \mathbb{F}^n$. We denote by $\delta^*((\mathbf{v}_1, \mathbf{v}_2), (\mathbf{w}_1, \mathbf{w}_2))$ the pairwise Hamming distance between two pairs, i.e.*

$$\delta^*((\mathbf{v}_1, \mathbf{v}_2), (\mathbf{w}_1, \mathbf{w}_2)) = |\{i \in [n] : \mathbf{v}_1[i] \neq \mathbf{w}_1[i] \wedge \mathbf{v}_2 \neq \mathbf{w}_2[i]\}|.$$

For linear code C, we denote by $\Delta^*_C(\mathbf{v}_1, \mathbf{v}_2)$ the following quantity.

$$\Delta^*_C(\mathbf{v}_1, \mathbf{v}_2) = \min_{(c_1, c_2) \in C \times C} \delta^*((\mathbf{v}_1, \mathbf{v}_2), (c_1, c_2))$$

Definition 7 (Foldable Code). *Let $c, d \in \mathbb{N}$ and for each $i \in [d]$, define $k_i = 2^i, n_i = c \cdot k_i$. A $[n_d, k_d]$, foldable code is a family of codes $(C_0, .., C_d)$, where the base code, C_0, is equal to the repetition code, $\{m^{||c} : m \in \mathbb{F}\}$ and each $[n_i, k_i]$ code, C_i, and each $\mathbf{c} \in C_i$ satisfies the following:*

$$\begin{aligned} \mathbf{c} := Enc_{C_i}(\mathbf{m}) = Enc_{C_{i-1}}(\mathbf{m}_L) + \mathbf{t}_i \circ Enc_{C_{i-1}}(\mathbf{m}_R) \\ ||Enc_{C_{i-1}}(\mathbf{m}_L) - \mathbf{t}_i \circ Enc_{C_{i-1}}(\mathbf{m}_R) \end{aligned}$$

where $\{\mathbf{t}_i \in \mathbb{F}^{n_i} : i \in [1, d]\}$ is given in the description of the code, $\mathbf{m} = (\mathbf{m}_L || \mathbf{m}_R)$ is a vector in $\mathbb{F}^{k_i}$, and $\circ$ denotes the Hadamard product.

Remark 2. In the above definition, elements $\mathbf{c}[j]$ and $\mathbf{c}[j + n_i/2]$ are two evaluation points on the polynomial $P(X) = \mathsf{Enc}_{C_i}(\mathbf{m}_L)[j] + X \cdot \mathsf{Enc}_{C_i}(\mathbf{m}_R)[j]$. For the remainder of the document, we assume wlog an ordering where these two elements are next to each other, i.e. where $\mathbf{c}[j], \mathbf{c}[j + 1]$ are two evaluations of the same polynomial.

Definition 8 (Fold). *Let $\mathcal{L}_{(x_0,y_0),(x_1,y_1)}$ be the lagrange polynomial for degree-1 univariate polynomial interpolation, i.e.*

$$\mathcal{L}_{(x_0,y_0),(x_1,y_1)}(X) = y_0 \frac{X - x_1}{x_0 - x_1} + y_1 \cdot \frac{X - x_0}{x_1 - x_0}.$$

For $l \in \mathbb{N}$, $\mathbf{v} \in \mathbb{F}^l$, $j \in \mathsf{even}(l)$, denote by

$$\mathcal{L}_{\mathbf{v},j} = \mathcal{L}_{(\mathbf{t}[j],\mathbf{v}[j]),(-\mathbf{t}[j],\mathbf{v}[j+1])}.$$

Furthermore, denote by $\mathsf{fold}(\mathbf{v}, r)[j]$ the following element.

$$\mathsf{fold}(\mathbf{v}, r)[j] = \mathcal{L}_{\mathbf{v},j}(r).$$

Let $A, B \in \mathbb{F}$ such that $AX + B = \mathcal{L}_{v,j}$. We denote by $\mathsf{unfold}(v, j)$ the pair of elements (A, B). Finally, we will sometimes fold over entire sets $S \subset [l]$, and this operation is well defined as long as S contains $j+1$ whenever it contains j. We denote this set-wise folding as $\mathsf{fold}(\mathbf{v}, r)[S]$ where,

$$\mathsf{fold}(\mathbf{v}, r)[S] = \{\mathsf{fold}(\mathbf{v}, r)[j] : j \in \mathsf{even}(S)\}.$$

2.5 The BaseFold IOPP

Let $d \in \mathbb{N}$. We review the BaseFold IOPP in Figure 1 for an $[n_d, k_d]$ foldable code, C, paramaterized by vectors $\{\mathbf{t}_i \in \mathbb{F}^{n_i} : i \in [1, d]\}$.

3 Folding Preserves Hamming Distance on Sets Within 1.5 Johnson Bound

In this section, we state and prove our main result. We fix constants $\epsilon, \eta, \beta \in [0, 1]$ such that $\beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, $\epsilon < 0.18$, and $\eta + 3\beta < 3$ and $\eta < \beta/3$, where these particular bounds are necessary to achieve our result while remaining realistic for our applications.

Definition 9 (Bad Coefficients and Sets). *Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{F}^n$. $r \in \mathbb{F}^*$ is a bad coefficient if there exists a set $S \subset [n]$ with more than βn elements such that $\Delta(\pi_L[S] + r\pi_R[S], C[S]) = 0$ but $\Delta^*_C(\mathbf{v}_L[S], \mathbf{v}_R[S]) > \eta n$. In this case, we refer to S as a bad set. We denote the set of bad coefficients by $\mathsf{BC}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$ and the set of bad sets by $\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$.*

Protocol 1 IOPP.commit

Input oracle: $\pi_d \in \mathbb{F}^{n_d}$
Output oracles: $(\pi_{d-1}, \ldots, \pi_0) \in \mathbb{F}^{n_{d-1}} \times \cdots \times \mathbb{F}^{n_0}$

- For i from $d-1$ downto 0:
  1. The verifier samples and sends $\alpha_i \leftarrow\!\!\$\ \mathbb{F}$ to the prover
  2. For each index $j \in \mathsf{even}[0, n_{i+1} - 1]$, the prover
     (a) sets $f(X) := \mathcal{L}_{\mathbf{v},j}(X)$
     (b) sets $\pi_i[j] = f(\alpha_i)$
  3. The prover outputs oracle $\pi_i \in \mathbb{F}^{n_i}$.

Protocol 2 IOPP.query

Input: *Oracles*: $(\pi_d, .., \pi_0)$, *Repetition Parameter*: $\lambda \in \mathbb{N}$

- Let $T = (V, E)$ be a binary tree with n leaves. For each $i \in [d+1]$, label the vertices in $V[i]$ by the elements of $[n_i]$ in order from least to greatest.
- For $j \in [0, \lambda - 1]$
  - The verifier samples an index $\mu_j \leftarrow\!\!\$\ [n_d]$
  - For each $i \in [d+1]$, with $(k, l) = P'_T(\mu_j)[i]$ queries the pair $\pi_i[k], \pi_i[l]$.
  - For i from d downto 1, with $k = P'_T(\mu_j)[i-1]$, the verifier checks that $\mathsf{fold}(\pi_i, \alpha_i)[P'_T(\mu_j)[i]] = \pi_{i-1}[k]$
  - If π_0 is a valid codeword w.r.t. generator matrix C_0, output accept, otherwise output reject

Fig. 1: BaseFold IOPP for foldable codes.

Theorem 3 (Relaxed Mutual Agreement Within 1.5 Johnson Bound). *Let C be a linear error-correcting code. For all $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{F}^n$,*

$$|\mathsf{BC}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}| \leq \frac{3}{\epsilon\eta} \tag{3}$$

Throughout this proof we will introduce helpful claims as needed. For ease of exposition, we defer these smaller proofs to the end of the main proof.

Proof. We show first that $|\mathsf{BC}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}| \leq |\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}|$. Looking ahead, this will enable us to focus our analysis on the size of $\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$.

Lemma 1.

$$|\mathsf{BC}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}| \leq |\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}|.$$

Proof. Define the function $f : \mathsf{BC}_{(\mathbf{v}_L,\mathbf{v}_L),\epsilon,\eta} \to \mathsf{S}_{(\mathbf{v}_L,\mathbf{v}_R),\epsilon,\eta}$ as follows. $f(r) = S$ if $S \subset [n]$ is the largest set such that $\Delta(\mathbf{v}_L[S] + r\mathbf{v}_R[S], C[S]) = 0$ but $\Delta^*_C(\mathbf{v}_1[S], \mathbf{v}_2[S]) > \eta n$. We argue that f is an injection. Let r_1, r_2 be two distinct elements in $\mathbb{F}$ and suppose that $f(r_1) = f(r_2)$. Then there exists $c_1, c_2 \in C$ such that

$$\mathbf{v}_L[S] + r_1\mathbf{v}_R[S] = c_1[S],$$

$$\mathbf{v}_L[S] + r_2\mathbf{v}_R[S] = c_2[S].$$

By linearity of the code, we subtract the bottom equation from the top equation and obtain

$$(r_1 - r_2) \cdot (\mathbf{v}_R[S]) = (c_1 - c_2)[S].$$

Again by linearity of the code, we divide both sides by $(r_1 - r_2)$, which proves that

$$\mathbf{v}_R[S] = \frac{(c_1 - c_2)}{r_1 - r_2}[S].$$

Therefore, $\mathbf{v}_R[S] \in C[S]$. Furthermore, plugging this back into Equation 1, we find that $\mathbf{v}_L[S] \in C[S]$, which leads to a contradiction and completes the proof.

Lemma 2.

$$|\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}| \leq \frac{3}{\epsilon\eta}$$

Proof. We start with the following claim.

Claim. The total number of triples, $S_1, S_2, S_3 \in \mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$ such that $|S_1 \cap S_2 \cap S_3| \geq (1 - \Delta_C)n$ is less than or equal to $\binom{|\mathsf{S}_{(\mathbf{v}_2,\mathbf{v}_2),\epsilon,\eta}|}{2} \cdot \frac{1}{\eta}$.

Proof. There are $\binom{|\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}|}{2}$ pairs of sets in $\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$. We show that each pair only intersects with $\frac{1}{\eta}$ other sets in $\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$. For the remainder of the proof, let $S_1, S_2 \subset \mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$.

Fact 1 *Let $S_{1,2} = S_1 \cap S_2$. There exists two codewords $c_L^*, c_R^* \in C$ such that $\mathbf{v}_L[S_{1,2}] = c_L^*[S_{1,2}]$ and $\mathbf{v}_R[S_{1,2}] = c_R^*[S_{1,2}]$.*

Proof. By definition of bad sets, we have the following two equations for some $c_1, c_2 \in C$ and $r_1, r_2 \in \mathbb{F}$.

$$\mathbf{v}_L[S_{1,2}] + r_1\mathbf{v}_R[S_{1,2}] = c_1[S_{1,2}]$$
$$\mathbf{v}_L[S_{1,2}] + r_2\mathbf{v}_R[S_{1,2}] = c_2[S_{1,2}]$$

By definition of bad set, $r_1 \neq r_2$. Therefore, setting $c_R^* = c_1 - c_2/r_1 - r_2$ and $c_L^* = c_1 - (r_1 \cdot c_R^*)$ completes the proof.

Fact 2 *Define $\mathcal{S}_{(S_1,S_2)} \subseteq \mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$ as the sets that intersect with $S_1 \cap S_2$ in more than $(1 - \Delta_C)n$ positions. Then*

$$|\mathcal{S}_{(S_1,S_2)}| \leq \frac{1}{\eta}.$$

Proof. For $S \in \mathcal{S}_{(S_1,S_2)}$, denote by $\mathcal{N}(S)$ the set $\{i \in S : \pi_L[i] \neq c_L^*[i], \pi_R[i] \neq c_R^*[i]\}$. Therefore

$$|\bigcup_{S \in \mathcal{S}_{(S_1,S_2)}} \mathcal{N}(S)| = \sum_{S \in \mathcal{S}_{(S_1,S_2)}} |\mathcal{N}(S)| \leq n,$$

where both the quality and inequality follow because by Fact 1, $\mathcal{N}(S_1) \cap \mathcal{N}(S_2) = \emptyset$. Furthermore, by definition of $\mathsf{S}_{(\mathbf{v}_1,\mathbf{v}_2),\epsilon,\eta}$, for each $S \in \mathcal{S}_{(S_1,S_2)}$, $|\mathcal{N}(S)| = \Delta^*_C(\mathbf{v}_L[S], \mathbf{v}_R[S]) \geq \eta n$. Therefore,

$$\sum_{S \in \mathcal{S}_{(S_1,S_2)}} |\mathcal{N}(S)| \geq \sum_{S \in \mathcal{S}_{(S_1,S_2)}} \eta n = |\mathcal{S}_{(S_1,S_2)}| \cdot \eta \cdot n,$$

and so, combining the previous two equations and since $\eta n > 0$,

$$|\mathcal{S}_{(S_1,S_2)}| \leq \frac{\sum_{S \in \mathcal{S}_{(S_1,S_2)}} |\mathcal{N}(S)|}{\eta n} \leq \frac{n}{n\eta} = \frac{1}{\eta}$$

We will use the following claim to bound the size of $\mathsf{S}_{(\mathbf{v}_L,\mathbf{v}_R),\epsilon,\eta}$. We defer its proof.

Claim. Let $\mathsf{S} \subset 2^n$ such that each $S \in \mathsf{S}$ has more than $\beta \cdot n$ elements and there are at most $\binom{|\mathsf{S}|}{2} \cdot \frac{1}{\eta}$ triples S_1, S_2, S_3 such that $|S_1 \cap S_2 \cap S_3| > (1 - \Delta_C)n$. Suppose further that for all $S_1, S_2, S_3 \in \mathsf{S}$, $|S_1 \cap S_2 \cap S_3| \leq \beta n$. Then,

$$|\mathcal{S}| \leq \frac{3}{\eta\epsilon}.$$

By definition of bad sets, each $S \in \mathsf{S}_{,\epsilon,\eta}(\mathbf{v}_1, \mathbf{v}_2)$ has more than βn elements. Furthermore, each triple of sets S_1, S_2, S_3 must satisfy $|S_1 \cap S_2 \cap S_3| \leq \beta n$, because otherwise $\mathbf{v}_L[S_1], \mathbf{v}_R[S_1]$ are in fact in $C[S]$. Therefore applying the preceding claim, it follows that

$$|\mathsf{BC}_{(\mathbf{v}_L,\mathbf{v}_R),\epsilon,\eta}| \leq |\mathsf{S}_{(\mathbf{v}_L,\mathbf{v}_R),\epsilon,\eta}| \leq \frac{3}{\epsilon\eta}.$$

Next, we prove the preceeding claim by genrealizing a proof of the Johnson boundp[4].

Proof (Proof of the preceding claim). We construct a bipartite graph, $G = (V, E)$, where right vertices are labeled by elements in $[n]$ and left vertices are labeled by sets in S. For $v_0, v_1, v_2 \in L$ and $w \in R$, define a "3-angle" as an 4-tuple, v_0, v_1, v_2, w, such that $(v_i, w) \in E$ for all $i \in [3]$. Let $d(w)$ be the degree of node w. Then the number of 3-angles in the graph is equal to $\sum_{v \in R} \binom{d(v)}{3}$. By Jensen's Inequality,

$$\sum_{v \in R} \binom{d(v)}{3} \geq |R| \binom{(\sum_{v \in R} d(v))/|R|}{3}. \tag{4}$$

Since each set in S has greater than or equal to βn elements, it follows that each vertex in L has more than βn neighbors. Furthermore, the sum of the degrees

[4] `https://www.cs.cmu.edu/~venkatg/teaching/au18-coding-theory/lec-scribes/list-decoding.pdf`

of all vertices in R is greater than $|L| \cdot \beta|R|$, i.e. $\sum_{v \in R} d(v) \geq |L| \cdot \beta|R|$. Thus, by Equation 4, it follows that,

$$\sum_{v \in R} \binom{d(v)}{3} \geq n \binom{\beta|L|}{3}.$$

Next, we prove an upper bound on the number of 3-angles in G. By assumption of the lemma, there are at most $\binom{S}{2}\frac{1}{\eta}$ triples, $S_1, S_2, S_3 \in \mathsf{S}$ such that $|S_1 \cap S_2 \cap S_3| > (1-\Delta_C)n$. Furthermore, $|S_1 \cap S_2 \cap S_3| \leq \beta n$. All other triples in S have three-way intersection less than or equal to $(1-\Delta_C)n < (\beta - \eta)^3 - \epsilon \leq \beta^3 - \epsilon$. Translating this to the graph G, we find that the total number of angles in G is at most

$$\binom{|L|}{3}(\beta^3 - \epsilon)n + \binom{|L|}{2}\frac{1}{\eta}\beta n.$$

Combining the two inequalities, we have,

$$\binom{\beta|L|}{3} \leq \binom{|L|}{3}(\beta^3 - \epsilon) + \binom{|L|}{2}\frac{1}{\eta}\beta.$$

Let $\alpha := \beta^3 - \epsilon$. Then,

$$\begin{aligned}
&\frac{\beta|L|(\beta|L|-1)(\beta|L|-2)}{6} \leq \frac{\alpha|L|(|L|-1)(|L|-2)}{6} + \binom{|L|}{2}\frac{1}{\eta}\beta \\
&\implies \frac{\beta|L|(\beta|L|-1)(\beta|L|-2)}{6} \leq \frac{\alpha|L|(|L|-1)(|L|-2)}{6} + \frac{|L|(|L|-1)}{2}\frac{1}{\eta}\beta \\
&\implies \beta|L|(\beta|L|-1)(\beta|L|-2) \leq |L|(\alpha(|L|-1)(|L|-2) + 3(|L|-1)\frac{1}{\eta}\beta) \\
&\implies \beta(\beta|L|-1)(\beta|L|-2) \leq \alpha(|L|-1)(|L|-2) + 3(|L|-1)\frac{1}{\eta}\beta \\
&\implies (\beta^2|L|-\beta)(\beta|L|-2) \leq (\alpha|L|-\alpha)(|L|-2) + 3(|L|-1)\frac{1}{\eta}\beta \\
&\implies (\beta^3|L|^2 - 2\beta^2|L| - \beta^2|L| + 2\beta) \leq \alpha|L|^2 - 2\alpha|L| - \alpha|L| + 2\alpha + \frac{3|L|\beta}{\eta} - \frac{3\beta}{\eta} \\
&\implies (\beta^3|L|^2 - 3\beta^2|L| + 2\beta \leq \alpha|L|^2 - 3|L|(\alpha - \frac{1}{\eta}\beta) + 2\alpha - \frac{3\beta}{\eta} \\
&\implies (\beta^3 - \alpha)|L|^2 - 3|L|(\beta^2 - \alpha + \frac{1}{\eta}\beta) + 2(\beta - \alpha) + \frac{3\beta}{\eta} \leq 0
\end{aligned}$$

where we derive each inequality by expanding, simplifying and/or rearranging terms. Let $P(X) = \epsilon \cdot X^2 - 3(\beta^2 - \alpha + \frac{\beta}{\eta})X + 2(\beta - \alpha) + \frac{3\beta}{\eta}$. We solve the inequality $P(|L|) \leq 0$ for $|L|$. Since P is concave, it is negative on the interval between its two roots, which we denote by x_0, x_1, with $x_0 < x_1$. We solve for x_1, which is an upper bound on L. By the quadratic equation,

$$x_1 = \frac{3(\beta^2 - \alpha + \frac{\beta}{\eta}) + \sqrt{(-3(\beta^2 - \alpha + \frac{\beta}{\eta}))^2 - 8\epsilon(\beta - \alpha + \frac{3\beta}{2\eta})}}{2\epsilon}$$

Next, we use the following claim to show that x_1 is a real root.

Fact 3 $x_1 \in \mathbb{R}$.

Proof.

$$\begin{aligned}
&(-3(\beta^2-\alpha+\frac{\beta}{\eta}))^2-8\epsilon(\beta-\alpha+\frac{3\beta}{2\eta}) \\
&=9(\beta^2-\alpha+\frac{\beta}{\eta})^2-8\epsilon(\beta-\alpha+\frac{3\beta}{2\eta}) \\
&>9(\frac{\beta}{\eta})^2-8\epsilon(1+\frac{3\beta}{2\eta}) && \text{(Because } \beta^2-\alpha>0 \text{ and } \beta-\alpha<1\text{)}
\end{aligned}$$

We have the following.

$$\begin{aligned}
&8\epsilon(1+\frac{3\beta}{2\eta})=8\epsilon(\frac{2\eta+3\beta}{2\eta}) \\
&<8\epsilon(\frac{2\beta+3\beta}{2\eta}) && \text{(Since } \eta<\beta\text{)} \\
&=8\epsilon(\frac{5\beta}{2\eta})<8\epsilon(\frac{6\beta}{2\eta})=8\epsilon(\frac{3\beta}{\eta}) \\
&<8\epsilon(\frac{\beta}{\eta})^2 && \text{(Since } 3<\frac{\beta}{\eta}\text{)} \\
&<9\epsilon(\frac{\beta}{\eta})^2 \\
&<9(\frac{\beta}{\eta})^2 && \text{(Since } \epsilon<1\text{)}
\end{aligned}$$

Therefore, since x_1 is a real root, we have,

$$\begin{aligned}
x_1 &\leq \frac{3(\beta^2-\alpha+\frac{\beta}{\eta})+\sqrt{(-3(\beta^2-\alpha+\frac{\beta}{\eta}))^2-8\epsilon(\beta-\alpha+\frac{3\beta}{2\eta})}}{2(\epsilon)} \\
&\leq \frac{3(\beta^2-\alpha+\frac{\beta}{\eta})+\sqrt{(-3(\beta^2-\alpha+\frac{\beta}{\eta}))^2}}{2(\epsilon)} \\
&= \frac{3(\beta^2-\alpha+\frac{\beta}{\eta})+3(\beta^2-\alpha+\frac{\beta}{\eta})}{2\epsilon} \\
&= \frac{2\cdot(3(\beta^2-\alpha+\frac{\beta}{\eta}))}{2\epsilon}=\frac{3(\beta^2-\alpha+\frac{\beta}{\eta})}{\epsilon} \\
&= \frac{3(\beta^2-(\beta^3-\epsilon)+\frac{\beta}{\eta})}{\epsilon} \\
&= \frac{3(\beta^2-(\beta^3-\epsilon))}{\epsilon}+\frac{3\beta}{\eta\epsilon}
\end{aligned}$$

Next, we use the following claim to bound the numerator of the left summand.

Fact 4 *Let* $\epsilon \in [0,1]$ *such that* $\epsilon<\beta^3$ *and* $\epsilon<0.18$. *Then,*

$$\beta^2-(\beta^3-\epsilon)<\frac{1}{3}.$$

Proof. First, the function $f(\beta) = \beta^2 - \beta^3$ has a maximum at $\beta = \frac{2}{3}$, where $f(\frac{2}{3}) \approx 0.148$. Therefore, since $\epsilon < 0.18$, it follows that

$$\begin{aligned}
&\beta^2 - (\beta^3 - \epsilon) \\
&= \beta^2 - \beta^3 + \epsilon \\
&< 0.148 + 0.18 \\
&= 0.32 < 1/3
\end{aligned}$$

Therefore,

$$|L| \leq x_1 \leq \frac{1}{\epsilon} + \frac{3\beta}{\epsilon\eta} \leq \frac{\eta + 3\beta}{\epsilon\eta} \leq \frac{3}{\epsilon\eta},$$

where the last inequality follows by the assumption that $\eta + 3\beta < 3$. This completes the proof.

4 Improved Soundness Proof For BaseFold IOP

We prove soundness of BaseFold IOP for evaluation of a multilinear polynomial at a random point. For the remainder of the section, for vector $\mathbf{v} \in \mathbb{F}^n$, denote $\mathbf{v}_L, \mathbf{v}_R$ as the unfolding of $\mathbf{v}$ (Definition 8).

Theorem 4 (Soundness of Basefold IOP For Evaluation At A Random Point). *Let $\lambda \in \mathbb{N}$ be a security parameter, let $d \in \mathbb{N}$ be the number of rounds in the IOP, $\pi_d \in \mathbb{F}^{n_d}$, $\beta > (1 - \Delta_C + \epsilon)^{1/3} + d\eta$ and $l \in \mathbb{N}$ such that $\beta^l \leq \mathsf{negl}(\lambda)$. Let $C_0, .., C_d$ be a family of foldable codes (Definition 7). Then, with probability greater than*

$$1 - \frac{3d}{\epsilon\eta|\mathbb{F}|},$$

over verifier randomness $(r_d, .., r_1) \leftarrow \mathbb{F}$ in the `commit` phase, and letting $\{\pi_i \in \mathbb{F}^{n_i} : i \in [d+1]\}$ be the corresponding oracles sent by the prover, either the verifier accepts with probability less than β^l or $\exists P \in \mathbb{F}[X_1, .., X_d]$ such that $Enc_{C_0}(P(r_d, .., r_1)) = \pi_0$ and $\Delta(Enc_{C_d}(P), \pi_d) < (1 - (\beta - d\eta))n_d$.

Remark 3. Using techniques from [20], it is straightforward to derive soundness for evaluation at a arbitrary point by combining it with the sumcheck protocol [17].

Proof. Let $T = (V, E)$ be the binary tree described in the verifier (Protocol 2). Let $\mathcal{P} \subset V$ be the set of augmented paths that satisfy the verifier checks. Since there are a total of $n_d/2$ augmented paths in T, it follows that each verifier sample is a Bernoulli trial with success probability $\frac{|\mathcal{P}|}{n_d/2}$. After l queries, the probability of acceptance is $(\frac{|\mathcal{P}|}{n_d/2})^l$. Therefore, if the verifier accepts with probability greater than β^l, then $(\frac{|\mathcal{P}|}{n_d/2})$ must be larger than β, and so $|\mathcal{P}| > \beta n_d/2 = \beta n_{d-1}$. For $i \in [d+1]$ let S_i be the vertices in augmented paths $\bigcup_{P \in \mathcal{P}} P$ that are in level i of V, i.e. the set $\bigcup_{P \in \mathcal{P}} P \cap V[i]$. We prove the following.

Lemma 3. *$|S_d| = 2|\mathcal{P}|$ and for all $i \in [d]$,*

$$\frac{|S_i|}{|V[i]|} \geq \frac{|S_{i+1}|}{V[i+1]}.$$

We defer the proof to the end. By the preceding lemma, and since $|\mathcal{P}| > \beta n_{d-1}$, it follows that for each $i \in [1, d]$, $|S_i| \geq \beta n_i$. Furthermore, by definition of the verifier checks,

$$\mathsf{fold}(\pi_i, r_i)[S_i] = \pi_{i-1}[S_{i-1}].$$

Throughout the proof, we will sometimes abuse notation and denote by P_i both the polynomial in $\mathbb{F}[X_1, .., X_i]$ and it's encoding (i.e. $\mathsf{Enc}_{C_i}(P_i)$).

Claim. For $i \in [d]$, and $r_0, .., r_i \in \mathbb{F} \setminus \mathsf{BC}_{(\pi_{i,L}, \pi_{i,R}), \epsilon, \eta}$, if $\pi_0[S_0] \in C_0[S_0]$, then there exists $P_i \in \mathbb{F}[X_1, .., X_i]$ such that $\Delta(\pi_i[S_i], P_i[S_i]) < i \cdot \eta \cdot n$ and $P_i(r_i, .., r_0)[S_0] = \pi_0[S_0]$.

Proof. We proceed with induction on i. In the base case, when $i = 0$, the claim follows trivially. Next, suppose the claim is true for $i < d$. Then there exists $S_i' \subset S_i$, with $S_i' > |S_i| - i \cdot \eta \cdot n_i$ such that $\pi_i[S_i'] = P_i[S_i']$ and $P_i(r_i, .., r_0) = \pi_0[S_0]$.

Since $\pi_i[S_i'] = P[S_i']$, it follows that $\mathsf{fold}(\pi_{i+1}, r_i)[S_{i+1}] = P[S_i']$. Therefore

$$\pi_{i+1,L}[S_i'] + r_i \pi_{i+1,R}[S_i'] = P[S_i'].$$

Since $r_i \notin \mathsf{BC}_{(\pi_L, \pi_R), \epsilon, \eta}$, it follows by Definition 9 and definition of fold (Definition 8) that $\Delta^*_{C_i}(\pi_{i+1,L}[S_i'], \pi_{i+1,R}[S_i']) < \eta \cdot n_i$. Therefore there exists a subset $S^* \subset S_i'$, with $|S^*| > |S_i'| - \eta n_i$ such that $\pi_{i+1,L}[S^*]$ is equal to some codeword $c_L \in C$ and $\pi_{i+1,R}[S^*]$ is equal to a codeword $c_R \in C$. This implies that $\Delta^*_{C_i[S^*]}(\pi_{i+1,L}[S^*], \pi_{i+1,R}[S^*]) < |S_i| - (|S_i'| - \eta n_i) = |S_i| - (|S_i| - i \cdot \eta \cdot n_i) - \eta n_i = (i+1) \cdot \eta \cdot n_i$. By definition of fold, this implies that $\Delta_{C_{i+1}[S_{i+1}]}(\pi_{i+1}[S_{i+1}]) < 2(i+1)\eta n_i = (i+1)\eta n_{i+1}$. It is therefore left to prove that c_L, c_R encodes polynomial $P_L, P_R \in \mathbb{F}[X_1, .., X_i]$ such that $P_L + r_i P_R = P_i$. We know that $\pi_i[S^*] = P_i[S^*] = c_L[S^*] + r_i c_R[S^*]$ and $|S^*| > |S_i| - (i+1)\eta n_i > \beta n_i - (i+1)\eta n_i$. Since $\beta - d\eta > (1 - \Delta_C)$, it follows that $c_L[S^*] + r_i c_R[S^*]$ and P_i are equal as codewords.

By Theorem 3, each $r_i \in \mathbb{F} \setminus \mathsf{BC}_{(\pi_{i,L}, \pi_{i,R}), \epsilon, \eta}$ with probabiliy greater than $1 - \frac{3}{\epsilon \eta |\mathbb{F}|}$. Therefore, taking a union bound for $i \in [d]$, $r_1, .., r_d \in \mathbb{F} \setminus \mathsf{BC}_{(\pi_{i,L}, \pi_{i,R}), \epsilon, \eta}$ with probability greater than $1 - \frac{3d}{\epsilon \eta |\mathbb{F}|}$, which completes the proof.

Proof (Proof of Lemma 3). For each $i \in [d]$, each sibling pair in S_i maps to a distinct element in S_{i+1}. First, by definition of augmented path, each pair of siblings in S_{i+1} have a distinct parent in S_i. Therefore $\mathsf{parent}_T(S_{i+1}) = |S_{i+1}|/2$. By definition of augmented path, $\mathsf{parent}_T(S_{i+1}) \subset S_i$, therefore $|S_i| \geq |\mathsf{parent}_T(S_{i+1})| = |S_{i+1}|/2$. Therefore,

$$\frac{|S_i|}{|V[i]|} \geq \frac{|S_{i+1}|}{2|V[i]|} = \frac{|S_{i+1}|}{|V[i+1]|},$$

where the last inequality follows since by definition of binary tree, $2|V_i| = |V_{i+1}|$ for all $i \in [d]$.

5 Concrete Parameters

In this section, we report the number of verifier queries needed for the new BaseFold soundness proof and we compare it to the number of verifier queries in the original BaseFold protocol, as well as to the concurrent result proving mutual correlated agreement for Reed-Solomon codes from [16]. Here we put aside the fact that the latter only works for Reed-Solomon codes. In Table 1, we do an apples-to-apples comparison between the results from [16] and the original BaseFold results, looking at soundness error and number of verifier repetitions across different distances. We find that our soundness error is considerably lower and the number of verifier repetitions range from equivalent to twice as many compared to [16]. Compared to BaseFold, our verifier repetitions are about 2 times fewer. Next, we demonstrate the tradeoff possible by varying the value for η in our BaseFold soundness proof. We find that by raising the value of η, we are able to lower the number of "bad" elements, which enables us to use a field that is 2^{13}-times smaller. Results are in Table 2.

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Distance	Soundness Error	Num Verifier Repetitions
Hab24 [16]		
1/2	2^{-50}	360
3/4	2^{-49}	129
7/8	2^{-47}	83
15/16	2^{-46}	58.5
31/32	2^{-44}	44
This result		
1/2	2^{-87}	348
3/4	2^{-87}	172.7
7/8	2^{-87}	119
15/16	2^{-87}	95
31/32	2^{-87}	83
Original BaseFold Result [20]		
1/2	2^{-87}	380
3/4	2^{-87}	241
7/8	2^{-87}	201
15/16	2^{-87}	185
31/32	2^{-87}	178

Table 1: We consider instance sizes of 2^{30} over finite field, $\mathbb{F}$ such that $\log_2(|\mathbb{F}|) = 128$. We report the number of Basefold verifier repetitions needed to achieve a soundness error in the query phase below 2^{-100}. For our result and the original BaseFold result, we set $\epsilon = 0.05$ and $\eta = 0$ (which reduces to regular mutual correlated agreement without loss). For [16], we set $m = 3$ to minimize it's soundness error.

η	**Num Bad Elements**	**Num Verifier Repetitions**	**Required Field Size**
Distance 3/4			
5×10^{-7}	2^{31}	172	2^{131}
5×10^{-5}	2^{25}	173	2^{125}
5×10^{-3}	2^{18}	348	2^{118}
Distance 7/8			
5×10^{-7}	2^{31}	119	2^{131}
5×10^{-5}	2^{25}	120	2^{125}
5×10^{-3}	2^{18}	201	2^{118}

Table 2: We consider instance sizes of 2^{30} over finite field, $\mathbb{F}$. We vary the value of η from Theorem 3. The "Required Field Size" column reports the required field size to achieve soundness error of 2^{-100}. We set $\epsilon = 0.05$

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